Fourfold Structural Template: Locked Lemmas Comparing Missing-Sector Defect Delta_miss and Hodge Defect Delta_Hdg

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ORCID: 0009-0006-7068-3718 | GitHub: BenFrohman/fourfold-structural-template
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Abstract

Standalone mathematical specification ('README.md', 'INTEGRAL_HODGE.md') proving the locked structural equivalence L <-> (Delta_miss = empty) for candidate cycle-spanning submodules on projective fourfolds, and establishing why L does not imply vanishing of the ambient Hodge defect Delta_Hdg = empty.

1 Scope, Scientific Honesty, and Non-Claim Boundary

Establishes logical boundary lemmas showing why local span closure L neither proves nor disproves the rational Hodge Conjecture.

- **Repository:** `https://github.com/BenFrohman/fourfold-structural-template`
- **Dedicated 1-to-1 Zenodo DOI:** `10.5281/zenodo.23652996`
- **Earliest Git Commit SHA:** `8daf20ee89b95ce3190f5cca3b4132469e5f574f` (2026-09-24 06:48:14 PM CDT (UTC-5))
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2 Technical Overview & Repository Specification

Locked fourfold lemmas: L iff Delta_miss empty; L does not imply Delta_Hdg empty. Not a Hodge disproof.

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3 Cryptographic Provenance & File Inventory

Every source artifact in `BenFrohman/fourfold-structural-template` is audited via `titan-audit` (`audit/TITAN_AUDIT_CERTIFICATE.json`) and sealed under a deterministic SHA-256 Merkle tree via `sealguard` (`PROVENANCE_MERKLE_ROOT.json`) signed with Ed25519 key fingerprint `b428fb72676f1c52`.

Repository Path	Size	SHA-256 Prefix
INTEGRAL_HODGE.md	1360 B	c6fe3e4500278836...
LICENSE	11355 B	9da72c2bee4f20ba...
NOTICE	272 B	1ab014bc1110a051...
README.md	877 B	958d715aa4dd190e...

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